Implementing a Windows Account Lockout Policy with Change Management

Security Control Implementation and Change Management | Proven Solution

Cybersecurity Project Portfolio

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1. Project Overview

This hands-on project demonstrates how I translated an access-management requirement into a Windows domain security control, documented the intended change, and tested its behavior in a controlled virtual lab. The goal was to reduce the opportunity for repeated password guessing, help with the control of brute force attack, against user accounts while showing how security configuration should be supported by policy, testing, and change-management records.

I built the lab in Oracle VirtualBox using Windows Server 2016, Active Directory Domain Services, Group Policy Management, and a Windows 8.1 client. The work brought together four pieces of evidence: an access management policy, the configured Group Policy settings, screenshots from account lockout tests, and a change record.

2. Security Problem and Objective

Repeated failed sign-in attempts can indicate password guessing or brute-force activity. The proposed control locks an in-scope account after five consecutive failed login attempts, limiting repeated online guesses. A temporary lockout can also disrupt legitimate users, so the design includes a defined duration, testing, and an administrative recovery process.

Account lockout threshold	**5 invalid sign-in attempts**
Account lockout duration	30 minutes
Reset account lockout counter after	30 minutes
Configuration location	Default Domain Policy → Computer Configuration → Policies → Windows Settings → Security Settings → Account Policies → Account Lockout Policy
Test environment	Windows Server 2016 domain controller and Windows 8.1 client in Oracle VirtualBox

3. My Role and Implementation Approach

I reviewed the access management policy, identified the intended control settings, configured the domain account lockout policy through Group Policy Management, and tested sign-in behavior using lab accounts. I also maintained configuration screenshots and a change record showing the purpose and requested implementation of each setting.

4. Deliverable A — Access Management Policy

The supplied "Access Management Policy — Proven Solution" is marked Draft for approval. It sets out the purpose, scope, safeguards, and sanctions. Its key safeguards require lockout after five failed attempts (AM-01), implementation through change management (AM-02), authorized testing and retained evidence (AM-03), event review where supported (AM-04), periodic review (AM-05), and documented exceptions (AM-06).

Policy status: proposed/draft. The project does not establish that a designated approver formally approved or published this policy.

5. Deliverable B — Configuration Evidence

The Windows Server screenshot below records the account lockout settings in the Group Policy Management Editor. It provides direct evidence of the configured values visible in the lab.

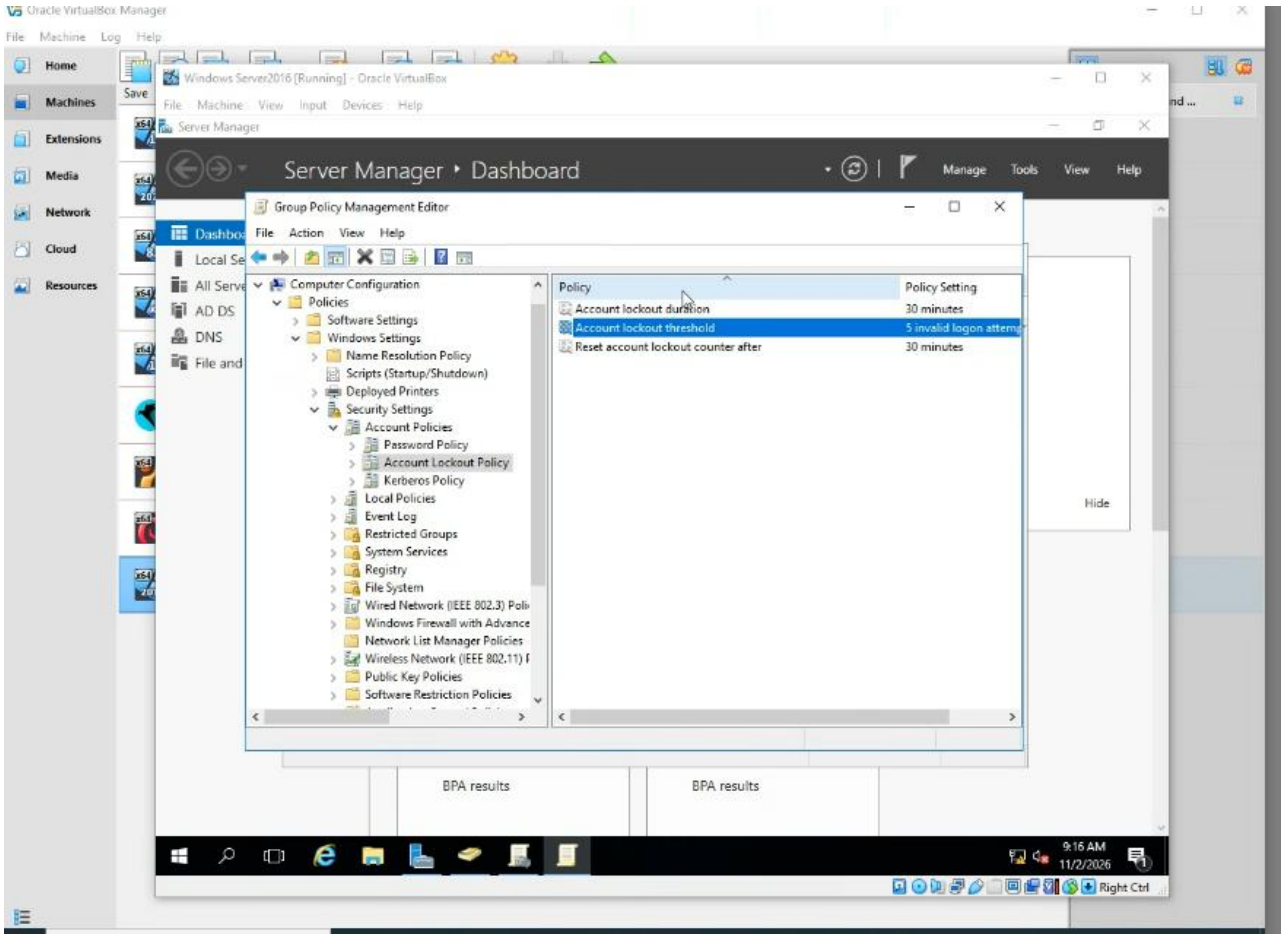


Figure 1. Account Lockout Policy settings: 30-minute duration, 5 invalid attempts, and 30-minute counter reset.

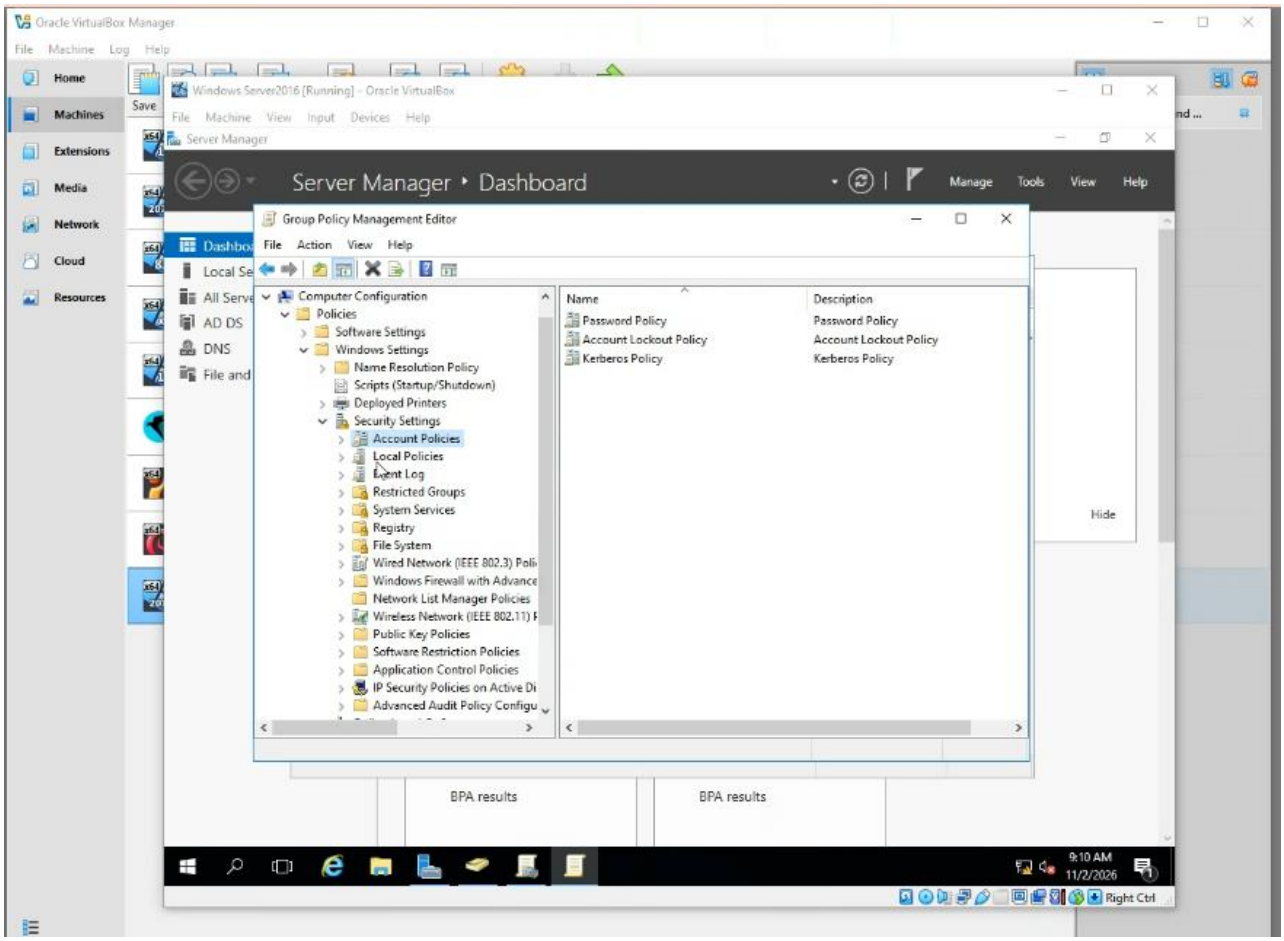


Figure 2. Group Policy Management Editor showing the Account Policies location in the Windows Server 2016 lab.

6. Deliverable C — Test Evidence

After applying the account lockout settings, I attempted to sign in to test accounts in the Windows 8.1 virtual machine. The screenshots show the system reporting that the referenced account was locked and could not be logged on. This demonstrates the lockout condition was encountered in the lab; the screenshots alone do not establish the exact number or sequence of failed attempts that preceded each lockout.

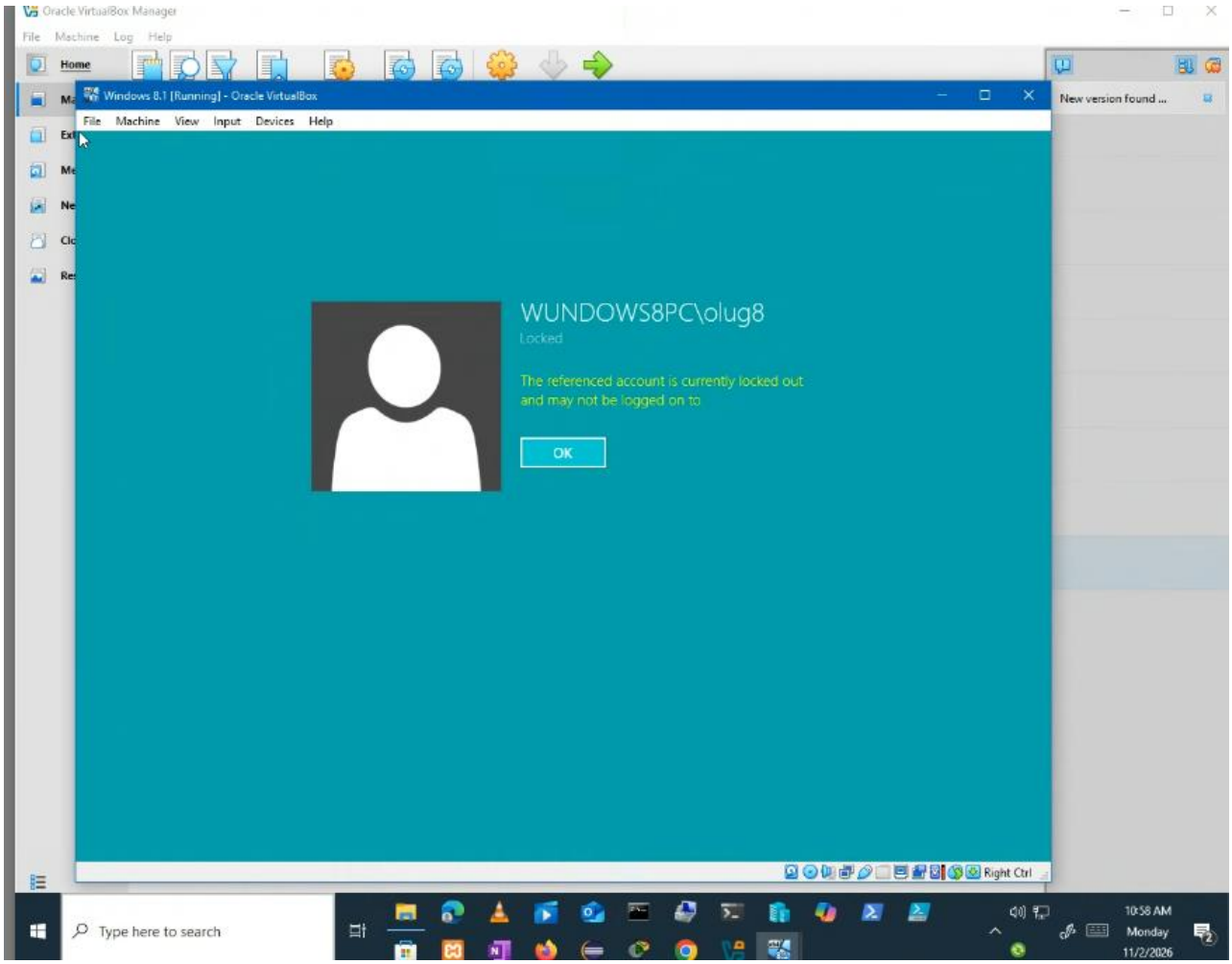


Figure 3. Domain user "user1" receives the account-locked message on the Windows 8.1 client.

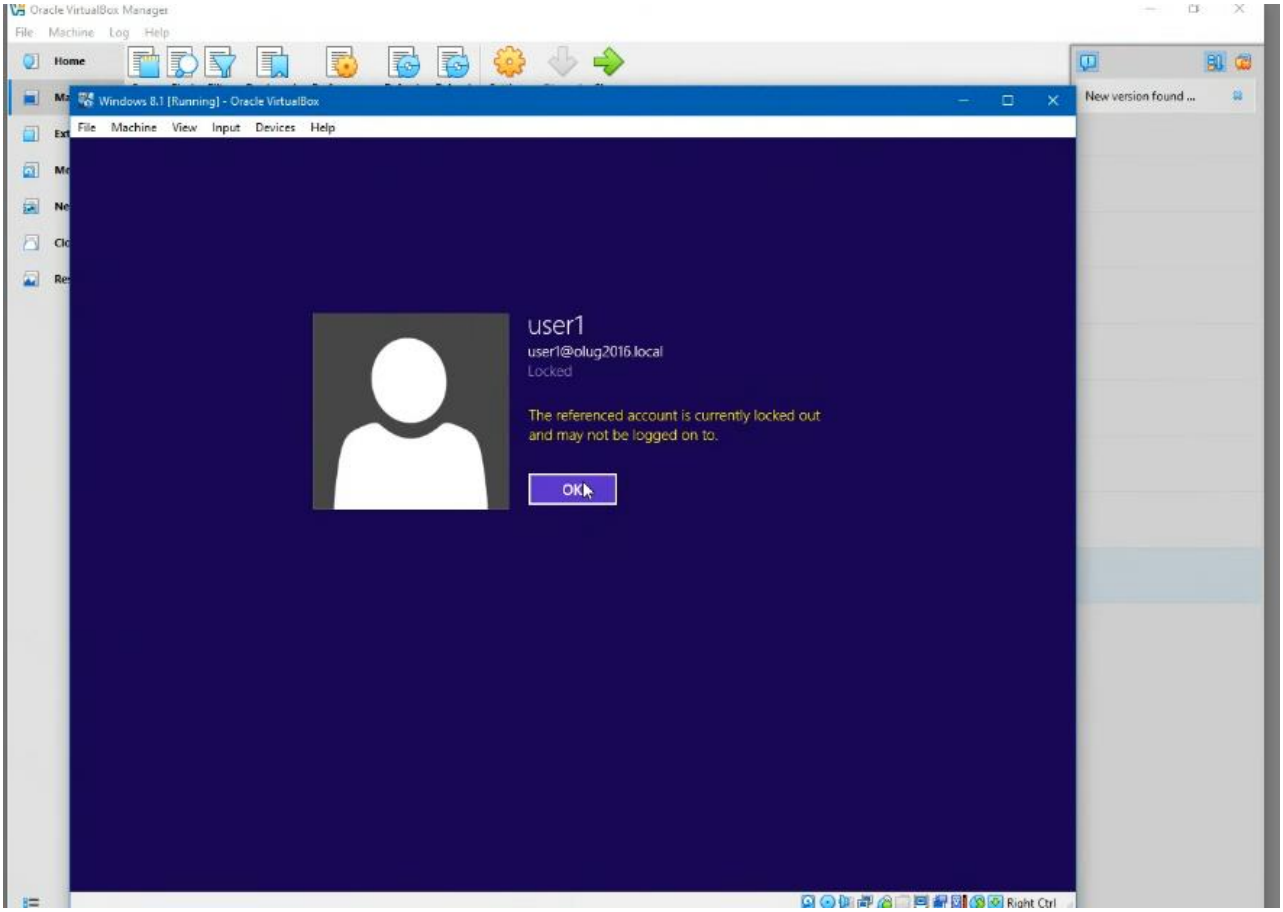


Figure 4. Local account "WINDOWS8PC\olug8" receives the account-locked message. This is separate from the domain-user example and does not, by itself, prove the domain GPO caused the local account lockout.

7. Deliverable D — Change Management Record

The supplied IT Infrastructure Change Record Log identifies Active Directory Domain Services as the affected system. All three requests are dated October 10, 2026, and carry an implemented-date entry of October 28, 2026. They are requested by Admin.Olaiya and categorized as GPO / Security. The record still lists each item as Pending Approval.

CHG-2026-001	**Set account lockout threshold to 5 attempts — reason: prevent brute-force credential attacks — Pending Approval**
CHG-2026-002	Set account lockout duration to 30 minutes — reason: standard temporary lockout window — Pending Approval
CHG-2026-003	Reset lockout counter after 30 minutes — reason: align observation window with lockout duration — Pending Approval

Change-control limitation: although the log includes implementation dates, approval is not evidenced. The supplied log also does not contain a completed risk/impact assessment, rollback plan, approval signature, or formal closure record. These remain documentation gaps rather than completed steps.

8. Results, Lessons Learned, and Security Relevance

The lab screenshots support that the specified account lockout values were configured and that lockout messages appeared during account testing. The exercise strengthened my ability to connect a written security requirement to a technical implementation, gather configuration and test evidence, and document changes in a traceable manner.

The control can help slow online password-guessing attempts, but it is not a complete defense against credential attacks. It should be paired with monitoring, strong authentication, appropriate password controls, and a process for restoring legitimate access. Lockouts can also be deliberately triggered to disrupt users.

9. Relevance to My Proposed NIW Endeavor

This project illustrates a practical method for improving cybersecurity readiness: establish a control requirement, implement a repeatable safeguard, verify system behavior, and maintain evidence for review. These are transferable skills for supporting organizations that need stronger identity and access controls, disciplined configuration management, and audit-ready security operations. As a simulated training project, it demonstrates technical capability rather than independently proving broad public impact or production outcomes.

10. Exhibit Index and Evidence Limitations

Exhibit A	**Access Management Policy — Proven Solution (user-supplied PDF; draft for approval)**
Exhibit B	Configuration screenshots — Group Policy Management and lockout values (Figures 1–2)
Exhibit C	Test screenshots — domain and local account lockout messages (Figures 3–4)
Exhibit D	IT Infrastructure Change Record Log — CHG-2026-001 through CHG-2026-003 (user-supplied PDF; pending approval)